



Fig. 7: High-speed DC electronic load from Chroma Instruments



Fig. 8: ASE Test's SSL 3.0 Test System (Image courtesy: www.ledtimes.com)



Fig. 9: Chroma's 58158-SC LED lighting inline test system

drivers and LED fixtures across multiple voltages and load capabilities in 10 to 20 seconds. Manufacturers can fulfil six-sigma certifications for their products with statistical process control (SPC) delivered by this system. To help construct industry standards, this testing system also creates defined testing parameters and benchmarks for LED light engines, drivers and systems.

A test system for LED bulbs, lamps and other luminaries

This LED lighting inline test system focuses on production-line tests. Apart from basic measurement, other processes such as data collection and analysis, integration of production line and automated calibration

Point of view: How PXI accelerates LED testing

Traditionally, multiple instruments were used for carrying out a single characterisation test. PXI provides a single platform with multiple instruments in a single chassis, which is capable of providing multiple kinds of inputs and acquiring multiple variations in outputs from device-under-test (DUT), thereby encompassing all necessary test procedures. For example, to test for ideal characteristics of an LED bulb, a particular voltage needs to be applied and the current flowing through the LED must be monitored, thereby plotting its IV characteristics. For this, an AC voltage generator or power supply and a separate measurement device are needed. We have replaced this approach of using two separate instruments for measurement with a single unit called source measurement unit (SMU).

The latest module, PXIe4141 SMU, which we came up with, contains two different technologies. One is called source adapt technology, which speeds up the complete characterisation test as it allows customisation of the SMU's response to any type of load.

The other technology is the capability of pulse power. SMUs are also capable of handling a very high power rating for a very low duration of time known as pulse power. Here, these source an amount of current, which is generally not under the specification, for a certain amount of pulse, and during that time, SMUs also complete the test in real-time. Therefore you can complete the overall test for a high power using an SMU that has low power capabilities in a short duration.

—**Avichal Kulshrestha, technical marketing engineer, National Instruments**

Pricing trends

Price trends for test equipment have been going up, but not by much. We fight to keep our pricing the same; however, our suppliers raise the price for components and sometimes we are forced to raise our own. I cannot speak for our competitors, but added features are not the cause for rising prices.

—**Tom Juel, inside sales representative, CSA Group/Orb Optronix**

highlight the advantages of using this system, in addition to optical and electrical T&M such as total luminous flux of LED lamps (lm), correlated colour temperature (CCT), colour rendering index (CRI), power factor (PF) and luminous efficacy (lm/W).

Spatial distribution and flicker measurement characterisation are two of its core techniques. By using the angle between the LED lamp and solar cells as well as the measured parameters from solar cells, it can calculate the distribution of the optical field of the LED lamp. It can also perform measurement of a flicker of an LED lamp with the help of high-speed response of solar cells.

This test equipment is suitable for different types of LED lamps. A special module is used to measure the luminous flux of inconsistent shapes or multiple sizes of LED lamps. This equipment is also easy-to-integrate with other automated systems to observe and increase test speed and reduce measurement cost to carry out automation on production line.

Sorting and binning LEDs with a spectral meter

In the industrial manufacturing of LEDs, based on colorimetric and photometric specifications, sorting and binning LEDs generates challenges in measurement parameters such as accuracy and speed. Apart from error-free synchronisation of the spectral meter with the existing system operation, it is highly important for the measurement device to accurately obtain results and transfer the data in the shortest time possible.

Gigahertz-Optik's BTS2048-VL spectral light and colour meter is best suited for online production applications for radiometric, colorimetric, photometric and spectral-radiometric measurements. It provides a speedy response time over a broad dynamic range using three processors and electronic shutter, coupled with input and output triggers for synchronised LED-binning applications. ●